# Section 1.1 — Sets and Numbers

Math 103: College Algebra

**Standard Number Sets**

For each number below, circle the symbol for each set the number belongs to.

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. $-\sqrt{7}$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 2. $-\frac{75}{5}$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 3. $\frac{1}{5}$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ |
| 4. $-6$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 5. $5$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 6. $2$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ |
| 7. $\frac{7}{3}$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 8. $\frac{21}{3}$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 9. $3.61$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ |
| 10. $\pi$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 11. $-4.0$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 12. $-4$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ |
| 13. $\sqrt{8}$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 14. $\sqrt{9}$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 15. $-4.\overline{54}$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ |
| 16. $-0.2$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 17. $-3215$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 18. $9.0$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ |
| 19. $816$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 20. $-\frac{10}{7}$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 21. $0.\overline{123}$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ |
| 22. $-\frac{3}{4}$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | 23. $0.1001110000...$ | $\mathbb{N}$ | $\mathbb{Z}$ | $\mathbb{Q}$ | $\mathbb{R}$ | | | | | |

Give an example of...

24. ... an integer that is not a natural number.

25. ... a rational number that is not an integer.

26. ... a real number that is not a rational number.

**Set Builder Notation**

For each set $A$ given below, list three elements of the set. Then, write the given number $x$ in a form that shows that $x \in A$.

27. $A = \left\{\frac{n}{8} \mid n \in \mathbb{Z}\right\}$ $\quad x = \frac{1}{2}$

28. $A = \left\{\frac{n}{d} \mid n \text{ is odd, } d \text{ is even}\right\}$ $\quad x = 0.75$

29. $A = \{x - 2 \mid x \in \mathbb{N}\}$ $\quad x = -1$

30. $A = \left\{\frac{1}{a} \mid a \in \mathbb{Q}\right\}$ $\quad x = 3$

**Properties of Real Numbers/Simplifying Expressions**

Simplify each expression below. Clearly show your work on a separate sheet of paper.

31. $2x + 3x$

32. $5a - 11a$

33. $2 + 5t - 3$

34. $2(x - 4)$

35. $4(t + 3) - 2t$

36. $\frac{2}{3}(6t + 15)$

37. $\dfrac{2x + 10}{4}$

38. $3(x + 6) - 4(x - 3)$

39. $\dfrac{x + 5}{2} + \dfrac{x - 3}{3}$

40. $5 - 3.5(x - 1)$

41. $2.8x - \frac{1}{2}(x - 3)$

42. $3(x - y) - 2(x - 3) + 4(y + 2)$

43. $2.4(x - 6) + 11(2.1x - 5.4)$

**Properties of Equality/Solving Equations**

Solve each equation below. Clearly show your work on a separate sheet of paper.

44. $4x - 3 = 17$

45. $11x + 8 = 4$

46. $5y - 7 = 2y + 13$

47. $2(y - 33) + 3(y - 2) = 8$

48. $9x + 12 - 3x - 2 = x + 8 + 5x$

49. $3a - 4a + 9 = -12a + 43 - 6a$

50. $4 - (6x + 5) = 7 - 2(3x + 4)$

51. $5(a - 1) - 9(a - 2) = -3(2a + 1) - 2$

52. $3(2x - 5) - 4x = 2(x - 3) - 12$

53. $\frac{1}{6}(m - 4) = \frac{2}{3}(m + 1) - \frac{1}{3}m$

54. $0.2 - 1.3t - 0.8t = 1 - 2.1t - 0.8$

55. $0.3(a - 18) = 0.9a$

56. $9 - 0.2(v - 8) = 0.3(6v + 8)$

57. $4.2y - 8.2 + 2.3y = 0.9 + 6.5y - 1.9$